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Co-producing gender knowledge in energy transitions: A grey literature perspective

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ABSTRACT

This paper explores how the issue of gender is co-produced within the context of energy transitions, and how gendered framings in grey literature actively shape both policy and research directions in the energy field. Drawing on Sheila Jasanoff's concept of co-production and building on critical examinations of energy and sustainability transitions within capitalist and Western frameworks, we analyse how gender is not simply added onto existing energy systems, but is entangled with the epistemic, institutional, and technological configurations that underpin those systems. Based on a purposive sample of 138 grey literature documents published between 2018 and 2023, we examine how gender is framed across energy subsystems (political, economic, socio-ecological, and technological) and across multiple levels of analysis—from household to global. Our findings show that while economic and socio-ecological concerns are commonly addressed, technological domains remain largely gender-blind. Gender is often reduced to "women", with limited intersectional analysis. Nevertheless, grey literature plays a key role in stabilizing certain visions of gender equality, contributing to the co-production of policy logics, intervention strategies, and research] agendas that define the contemporary energy transition.

1. Introduction

The sociological understanding of transition refers to the time period when crucial decisions are being made to shape the particular future (Wallis, 1970). The energy transition; often framed as a grand socio-technical transformation; carries with it not only the promise of decarbonizing infrastructures but also of reconfiguring social relations toward greater justice and equity. However; recent critical scholarship has questioned the assumption that sustainability transitions can unfold smoothly within capitalist or Western modernist frameworks. Scholars such as Feola (2020) argue that the dominant transition narratives often reproduce the very socio-economic logics that have generated ecological and social crises; leading to what he terms "capitalist realism" in sustainability transitions. Similarly; Velasco-Herrejón et al. (2022) highlight how the hegemonic discourse of sustainable development and ecological modernization continues to marginalize non-Western worldviews. Their analysis of Indigenous Zapotec communities in Mexico demonstrates that renewable energy projects; while framed as environmentally progressive; can reproduce colonial relations of dispossession and epistemic dominance. These insights underscore the need to view energy transitions not only as technological and governance processes but also as sites of contestation over knowledge; justice; and world-making. This perspective addresses both environmental concerns; but also systemic power asymmetries embedded within existing energy regimes; where inequalities constitute a crucial dimension of a just transition. Among these; gender inequalities have gained growing attention as a crucial dimension of a just transition. As the energy transition carries the risk of exacerbating existing gender gaps in the interrelated areas of energy governance; labour market; consumption and safety; there is a growing recognition of the need for mainstreaming gender into energy policies; programs; and projects to ensure that both women and men benefit equally from the transition to sustainable energy systems (Lahiri-Dutt, 2023). By linking the energy–gender nexus to these broader critiques of sustainable transition, our analysis seeks to illuminate how power and normative assumptions shape whose visions of "sustainable" futures are legitimized and whose are silenced.

To explore this, we draw on the concept of co-production developed by Sheila Jasanoff (2004), which posits that scientific knowledge and socio-legal order are symmetrical and mutually constitutive. From this perspective, energy systems are not only materially configured but also shaped by societal visions, legal frameworks, and institutional

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arrangements—just as those arrangements are stabilized and legitimated through specific forms of knowledge and expertise. The co-productionist framework thus allows us to see energy transition not as a purely technical process supplemented by social considerations, but as a deeply intertwined political and epistemic transformation.

This approach is particularly well-suited to analysing grey literature, which plays a crucial yet underexplored role in shaping the energy-gender nexus. The documents produced by international organisations, NGOs, think tanks, and policy actors working on promoting the use of renewable energy solutions reveal how gender expertise is integrated into, or omitted from, energy discourse, and how such framings reflect and reproduce wider institutional logics and governance arrangements. In this paper, we use co-production as a critical tool to examine how gender knowledge and normative visions of fairness are produced alongside energy technologies and policies. Based on a grey literature review, we analyse, how gender is represented, which categories and framings dominate (e.g., "women" as beneficiaries vs. "gender" as a structural axis), and how institutional contexts—European, African, and global—shape these representations. Europe has developed institutional frameworks and tools for monitoring gender and energy policy, including dedicated research initiatives and efforts to collect and analyse gender-disaggregated data (European Economic and Social Committee, 2024). However; dedicated gender-energy monitoring frameworks are emerging rather than fully established and there is still a notable lack of standardized gender-disaggregated data across many key energy policy indicators (European Institute for Gender Equality, 2016). Furthermore, to fully understand the dynamics of the gender–energy nexus, it is essential to extend the analysis beyond Europe and incorporate perspectives from diverse regional and institutional contexts where innovative approaches to integrate gender, intersectionality and indigenous knowledge into energy and climate policies have been pronounced. This broader scope allows for a more comprehensive understanding of how global approaches to energy and gender intersect, differ, and inform one another.

In particular, we attempt to answer the following research questions:

- How do non-governmental organisations and policy initiatives in Europe and worldwide frame the gender-energy nexus across different technologies, resources, and levels of analysis—from household to structural scales?
- What policy approaches and areas of intervention are proposed to address gender inequalities in energy systems globally, and how does the European experience compare?

Through this analysis, we aim to uncover how grey literature both reflects and participates in the simultaneous making of knowledge and social order, and how this affects the vision of gender justice embedded in energy transitions. This theoretical approach not only allows us to identify omissions and partial framings in existing discourse but also opens up space to imagine more inclusive and reflexive energy futures. It is increasingly acknowledged that part and parcel of these visions is gender equality in different energy subsystems (political, economic, socio-ecological, and technological).

Research on gender-energy nexus is systematically growing, evidencing gender inequalities in various areas of energy transition (Streimikiene, 2023; Castán Broto, 2024; Cellini et al., 2025; Henriques et al., 2025). However, peer-reviewed literature does not cover the full breadth of existing views and knowledge on the gender-energy nexus (Yoshida et al., 2024). Provided that gender dimension in the energy transition is a relatively novel topic; grey literature might cover relevant information not yet covered in the peer-reviewed literature (Van der Merwe et al., 2020); including insights into the ideas and practices of policy relevance (Swarnakar and Singh, 2022). It is also more accessible than academic publications and relevant to a wider range of readers.

2. Analysing gender-energy nexus: theoretical approach, grey literature research gap, and role of international organisations in shaping public knowledge

2.1. Co-production of gender-energy nexus in grey literature

Co-production is a concept that can be interpreted and applied in multiple ways. While stemming from different traditions, conceptualisations of co-production are interconnected and reflect how researchers and non-academic actors engage in knowledge production in response to complex global challenges (Bandola-Gill et al., 2023). For the sake of our paper; we follow the concept of co-production; as developed by Sheila Jasanoff; which offers a powerful lens to understand how knowledge and social order are formed simultaneously (Jasanoff, 2004, 2005, 2010; Bandola-Gill et al., 2023). Central to this approach is the idea of symmetry, which describes how scientific knowledge (facts, expertise, technologies) and socio-legal order (institutions, regulations, norms) are not hierarchically structured but are equally powerful and mutually constitutive forces (Jasanoff, 2004; Forsyth, 2019). Following that, we might assume that the design of energy systems is co-produced through normative visions, societal institutions, and legal frameworks, which are stabilized and legitimized by specific forms of expertise and technical knowledge.

Building on Feola (2020) and Velasco-Herrejón et al. (2022), who critically examine sustainability transitions within capitalist and Western frameworks, we approach these dynamics through a co-productionist lens that foregrounds how epistemic and normative orders are jointly constituted. This theoretical perspective is particularly useful for analysing the gender–energy nexus in grey literature. Policy documents and non-academic publications are more than passive carriers of information—they actively demonstrate how knowledge and social order are co-produced in the political realm. Grey literature may reveal how gender expertise is either integrated into or excluded from energy discourses, and how these choices shape institutional norms, policy architectures, and funding priorities. In this paper, we apply the co-productionist framework to examine how the gender–energy nexus is made visible—or rendered invisible—across EU, African, and global institutional contexts. This enables us to trace how energy governance is tied to specific epistemic assumptions about gender, justice, and sustainability, and how institutional framings may reproduce or contest those assumptions.

By focusing on grey literature, we can identify how gender is positioned in relation to technological systems, what forms of gender knowledge are considered legitimate, and how it is constructed. Rather than treating gender as an external variable, co-production helps us see it as constitutive of how energy systems are imagined, designed, and regulated. Importantly, this approach enables us to uncover the strategic omissions and framing choices in these documents—what categories are used (e.g., "women" vs. "gender"), which actors are given authority, and which dimensions of justice are prioritized or sidelined. It also helps us observe how technical discourse (about infrastructure, renewables, or innovation) intersects with legal and institutional norms (such as gender quotas, inclusion frameworks, or funding conditions), producing particular visions of energy transition. In short, co-production enriches our analysis by offering a critical tool for mapping the simultaneous formation of knowledge and normative order, and for understanding the epistemic and political consequences of these entanglements in shaping the future of energy transitions.

Grey literature has been infrequently used in reviews as evidence for reconstructing knowledge and approaches related to gender aspects of the energy transition. When it is included, it typically appears only as a supplement to peer-reviewed publications (Rewald, 2017; Winther et al., 2017; Pueyo and Maestre, 2019; Van der Merwe et al., 2020; Gender CC-Women for Gender Justice EV, 2024). These reviews generally demonstrate an insufficient and selective integration of gender perspective into scientific; policy and public reflections on energy

transition. For example; Pueyo and Maestre (2019) included grey literature in their review of publications linking energy access; gender; and poverty. In examining gender differences in the benefits derived from the Productive Use of Electricity (PUE); they found that the existing PUE literature had considered gender only at the household level; focusing primarily on the labour-supply effects of access to electricity. Similarly; Van der Merwe et al. (2020) incorporated grey literature in a structured review of publications on gendered energy innovations in urban poor environments. Their analysis also revealed gaps in the mainstreaming of gender perspective in energy innovations across policy, organisational, technological, household, and cultural levels of discourse.

One of the two identified separate grey literature reviews, analysing the EU reports and policy documents, also reported significant gaps demonstrating a disconnect between gender equality and clean energy strategies: while the EU's gender equality strategies are ambitious, major climate and energy policies remained largely gender-neutral or gender-blind, overlooking differentiated impacts and opportunities (Carroll, 2022). The other one; with a global focus; identified more comprehensive approaches by exploring existing political narratives around a gender-just transition (Walk, 2024). It worked out six narratives; each reflecting a specific feminist theory and underlying idea of gender justice. Two narratives are rooted in liberal feminism aiming at achieving gender equality within the framework of the current economic and social system through ensuring equal representation in politics and society. In line with this approach; the representation narrative focuses on increasing women's participation in energy policy-making; and the opportunity narrative investigates the ways of removing barriers for equal participation in green energy labour market and education. Three narratives share assumptions of socialist feminism viewing gender as structural; i.e.; embedded in social power and economic systems and post-colonial feminism; which focuses on intersectional inequalities caused by colonialism and neocolonial economic structures. Thus; the policy design narrative calls for integrating gender mainstreaming and intersectional analysis into all climate policies; the fossil phase-out narrative underlines the need for holistic regional transitions with social protection for informal and care-sector workers; and the protection narrative concentrates on strengthening women's access to resources; social protection; and public care systems. Only one narrative – the transformation narrative – assumes that a gender-just transition requires a radical systemic change; because capitalism; neocolonialism and patriarchy produce intersectional inequalities and exploitation. By recognising that these deep-seated structures need to be changed; and the policy of degrowth should be implemented; it synthesizes the assumptions of ecofeminism; socialist; and post-colonial feminism. At the same time however; the analysis demonstrates that postmodern feminism; which criticizes policies based on the single universal category of "women" for not considering gender and sex as shifting and fluid categories; is rarely represented in the gender-just transition discourse (Walk, 2024). By proposing a multi-layered feminist vision of a gender-just transition combining incremental reforms in representation and policy design with transformative agendas of degrowth and care economies; the author of this review aligns with a broader critical discourse on sustainability transitions represented by above mentioned research (Feola, 2020; Velasco-Herrejón et al., 2022). This approach is further developed by Martin et al. (2024) who propose a pluralist understanding of both sustainability and values—recognizing multiple legitimate pathways toward sustainability but also asserting that some values are more supportive than others. These include relational and justice-oriented values of stewardship, care, and responsibility; the values suppressed by colonial, patriarchal and capitalist dominance emphasising unsustainable values of individualism and profit maximization. Similarly, as in the transformation narrative of a gender-just transition, sustainability transition therefore requires addressing power asymmetries, especially those based on gender, class, and coloniality.

Grey literature is often absent from reviews of the gender–energy nexus due to the significant effort required for its retrieval (Johnson et al., 2020) and its frequent lack of methodological robustness and transparency; as documented by Feenstra and Özerol (2021) and Lazoroska et al. (2024). While these concerns are valid, excluding grey literature creates a significant gap, as public, and especially international, organisations play a key role in shaping imaginaries of a just energy transition.

2.2. The role of international organisations in shaping public knowledge on energy

Two major intergovernmental initiatives were selected to guide the identification of grey-literature sources for this study: the International Energy Agency (IEA) Technical Collaboration Programme *Equality in Energy Transitions* (EiET) and the United Nations' 2030 Agenda for Sustainable Development, which holds universal significance across all sectors of sustainable development, including energy. These overarching frameworks were chosen for their global governmental mandates and their central role in shaping and coordinating international policy discourse on gender and energy. The UN 2030 Agenda also encompasses the mandate and activities of relevant UN bodies such as UN Women, which plays a key role in advancing gender equality across sustainable development sectors, including those linked to energy. Moreover, these two initiatives represent the most up-to-date entry points for the international networks active in the gender–energy field.

At the international level, the EiET Initiative is one of the longest-standing initiatives paving the way for the recognition of the crucial role of women in the transformation to a clean energy economy. Over the years, this initiative has continually refined and deeply examined its main focus, which is now centred on "Accelerating gender equality and diversity in clean energy transitions." EiET actively promotes discussions in high-level fora, such as the joint annual Clean Energy Ministerial (CEM) and Mission Innovation (MI) and encourages the consideration and implementation of Gender, Equity, Diversity, and Inclusion (GEDI) principles in the delivery of CEM and MI activities. In addition, the EiET has become a well-recognized catalyst for networking among many organisations engaged in closing the gender gap in clean energy and accelerating the transition to a low-carbon economy. It offers a unique online platform for international collaboration, networking, and dissemination.

The UN 2030 Agenda is the global framework through which, in 2015, the United Nations charted the course to address the most urgent challenges of our time through collaborative partnerships among all stakeholders and to take the bold and transformative steps urgently needed to shift the world onto a sustainable and resilient path. The challenges needing urgent solutions are framed as Sustainable Development Goals (SDGs) and include gender equality (SDG5) and affordable and clean energy (SDG7). The review of the SDGs at the United Nations High-Level Political Forum in 2022 identified insufficient addressing of energy interlinkages with gender equality, mirrored in gender-blind energy policies and planning, which contributes to perpetuating gender inequalities in energy transition (UN (United Nations), 2022).[1]

Grey literature—produced by NGOs, international institutions, think tanks, and policy actors—plays a pivotal role in shaping public and policy discourses. Often identifying emerging societal issues before they reach academic research (Auger, 1998; Schöpfel, 2010), it provides context-rich, sector-specific, and locally grounded insights that inform theoretical development. Beyond highlighting new challenges, grey literature reveals policy gaps and evaluates the social impacts of implemented measures (Mahood et al., 2014). Its practice-oriented and

[1] In response to these findings, the IEA's Gender and Energy Data Explorer (https://www.iea.org/data-and-statistics/data-tools/gender-and-energy-data-explorer) was launched in 2022 to provide evidence-based analysis and support policy-makers in addressing gender disparities in the energy sector.

accessible format—designed for immediate applicability rather than academic rigor—enhances its influence among policy-makers; opinion leaders; and media; thereby bridging expert knowledge and public discourse to support more inclusive policy-making (Adams et al., 2016).

Unlike peer-reviewed "white literature," grey literature is heterogeneous and lacks standardized validation or replicability processes (Adams et al., 2016). Its credibility often depends on the issuing organisation; and its tendency to promote specific solutions can limit critical engagement. Inconsistent metadata and indexing further heighten the risk of omissions. Nevertheless; grey literature effectively complements recent scientific research across key domains—empowerment; employment; attitudes and behavior; transition; awareness; perception; and health (Cellini et al., 2025).

3. Methodology

The Grey Literature Review (GLR) presented here seeks to elucidate how the energy–gender nexus is conceptualized by organisations engaged in policy-making for sustainable transition. Adopting a purposive, non-probability sampling design, the review targets institutions with demonstrable influence in energy and gender policy discourses, prioritizing the capture of institutional perspectives over an exhaustive publication inventory.

3.1. Organisational scope and classification

The review sought to understand how global and regional organisations with different mandates and expertise in gender analysis conceptualize the energy–gender nexus. In the initial phase, organisations were identified and categorized by their scope and area of activity. A distinction was made between global and African organisations and those with a European focus, the latter including:

- Entities associated with the EU through research or technological development (e.g., the European Commission Research Programmes, ETIPs, and EU Energy Initiatives), and
- Organisations collaborating with EU bodies in policy-making but not institutionally affiliated, such as energy and gender NGOs.

The specific focus on African, and European organisations in the grey literature review reflects both the project's mandate and established patterns in the research field. The Horizon Europe gEneSys project, under which the review was conducted, was designed to focus on Europe and Africa, in line with the European Parliament's call *Towards a renewed partnership between Africa and the EU*, making organisational outputs from these regions directly relevant. Moreover, in the most comprehensive recent scoping review of academic publications on the gender–energy nexus (Cellini et al., 2025), studies on Sub-Saharan Africa formed the largest subsample, underscoring Africa's centrality in current scholarship. The inclusion of global organisations further ensured coverage of cross-regional evidence and international policy guidance, providing a balanced and contextually grounded foundation for the review.

This classification resulted in a matrix encompassing organisations of varied types, activity scopes, and interests (Table 1).

3.2. Search execution and data processing

Between July and September 2023 six research groups, each aligned with one of the organisational categories, systematically searched and analysed publications according to their respective areas of expertise. Each group employed the agreed-upon keywords, which included "gender & energy," "women & energy," "female & energy," and "girls & energy" and adapted its search strategy to the nature of the sources under review. For example, EU Research and Innovation projects were identified through the Cordis database and through the analysis of project deliverables and reports hosted on project websites. EU Energy Initiatives were located via Google searches and refined through targeted screening of materials from the EU Directorate-General for Energy website.

Documents were screened for eligibility based on the inclusion and exclusion criteria. We included reports, working papers, policy briefs, white papers, recommendations, and strategies published between 2018 and 2023, explicitly addressing both gender and energy transformation. Publications with only cursory mentions of gender or those framed purely within socio-demographic discussions, as well as non-policy materials such as newsletters, conference proceedings, presentations, theses, or event agendas were excluded. Initially, 232 publications were identified. Following screening for relevance and the application of inclusion and exclusion criteria, 138 publications were retained for analysis (Fig. 1).

3.3. Coding and data extraction

The data extracted from the comprehensive corpus were systematically categorized according to a series of analytical frameworks that were specifically aligned with the primary research objectives and the guiding questions posed (comp. Table 2). To maintain consistency and coherence in the coding process, three structured workshop sessions were convened at pivotal stages of the project: the initial workshop focused on aligning coding categories, the pilot workshop aimed at verifying the consistency of the coding, and the final workshop was dedicated to the review and discussion of the results. These sessions facilitated a unified understanding among the coding teams regarding the application of coding categories and the methodological approach to the data.

4. Results

4.1. Framing the energy-gender nexus in Europe and beyond

Given the transformative interplay between energy systems and socio-economic structures, the intersection of gender and the energy transition is crucial, with Jasanoff's concept of co-production offering a framework to explore how gender knowledge and institutional structures constitute and are constituted by energy systems. The reviewed publications address gender within the energy discourse to a varying extent (Fig. 2). Nearly half of publications position gender as the central analytical category, primarily exploring its role in energy systems, both traditional and renewable (51 publications), while others considering its broader intersection with climate (7 publications) or multi-sectoral domains, including but not limited to energy (8 publications). These publications apply gender perspective as a lens through which institutional, technological, and social transformations are interpreted and

Table 1
Typology of organisations included in the grey literature review, classified by geographic scope and policy focus.

Types of organisations	Worldwide and African organisations		Entities associated with the EU		Organisation collaborating with the EU in policy-making	
Organisational focus	General Focus	Energy Focus	Research team realizing EU's R&I programme	EU Energy initiatives and agendas	Gender Focus	Energy Focus
Items	74	23	27	6	5	3

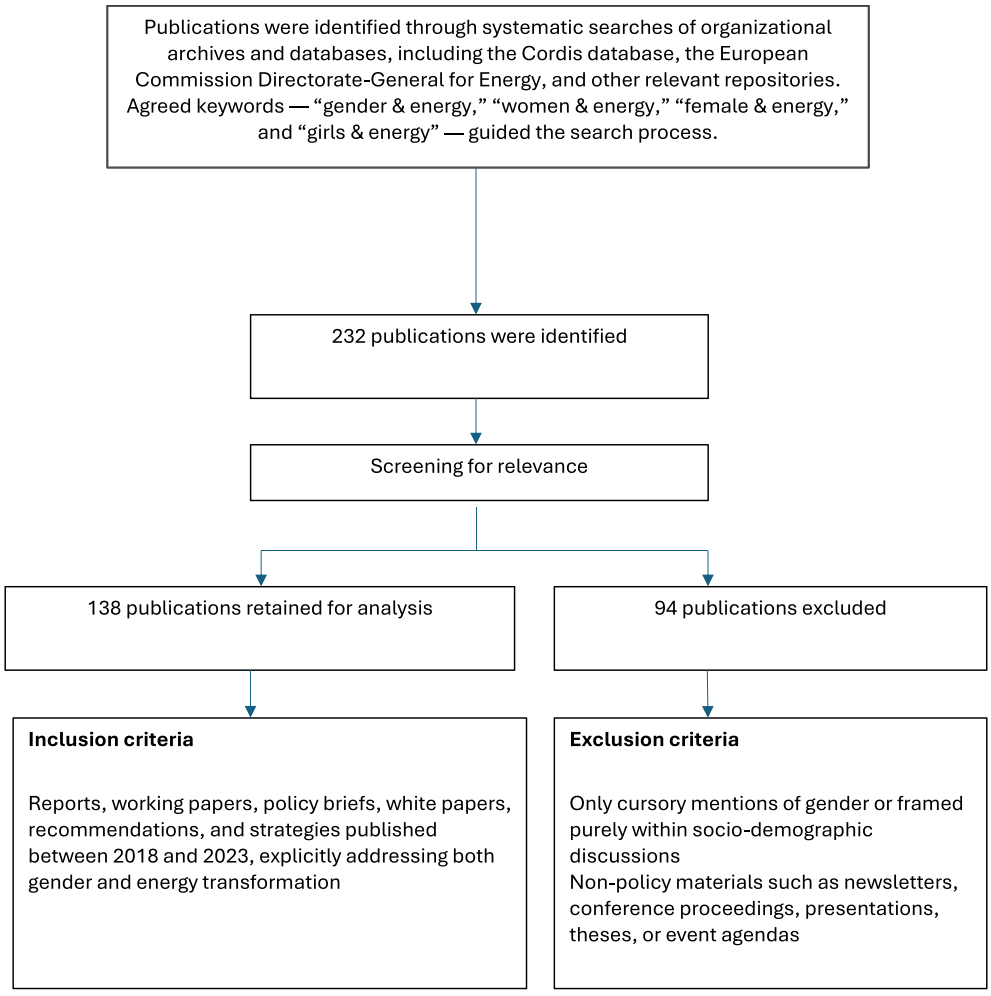


Fig. 1. Workflow summarizing the identification, screening, and inclusion process for the Grey Literature Review (2018–2023).

potentially restructured.

In publications where gender is the main focus, it is conceptualised as an analytical tool to expose how gender norms are institutionalised across domains such as climate, economy, and health, from organisational practices to macro policy levels (we2; ag1; ag7; eur5; wg38; wg39; wg47). These publications advocate for mainstreaming gender in energy transition, demonstrating its influence on institutional priorities, governance frameworks, and policy responses, often link it to SDGs, particularly gender equality as a precondition for a just energy transition (wg28).

Several publications where gender is the main theme highlight women underrepresentation in employment and leadership within the energy sector, especially in developing countries (we4; epg1; wg10). Women are often depicted as disproportionally affected by structural inequalities in energy access and participation, and increasing their presence is seen as beneficial both individually and societally.

A final frame visible in the publications where gender is analysed as an important perspective concerns gender-responsive policy (eur6; epg5wg2; eur14). While this is not a dominant theme, some studies argue for gender-transformative policies, and the inclusion of women lived experiences in energy governance. They point to the dialogue between knowledge, social order and policy-making by revealing how normative understandings of gender regime are reproduced in the energy system and advocating for restructuring the existing socio-technical systems. These perspectives fit well with Jasanoff's framework, by demonstrating how scientific practices, institutional priorities, and gender order are co-produced, and reinforce the need for structural transformation in the energy system.

In 52 publications, gender is discussed among other themes, mainly in relation to operational guidance and cooking. The former offers strategic recommendations to reduce inequality through empowerment and inclusion and provide a more complete understanding of current challenges (wg60; wg18; wg8). Yet these guidelines and recommendations lack a critical perspective on structural power balances. The latter frames cooking as a gendered practice, calling for a shift from purely technical solutions to a more inclusive socio-ecological approach that addresses broader goals such as health, development, and gender equality (wg65;wg16; wg26). Modern cooking technologies are not only seen as solutions to reduce pollution, labour burdens, and enable economic development, but as entry points for transforming societal structures. These publications reflect co-production by combining technocratic interventions and normative commitments.

The smallest subset of 20 publications includes gender as a variable without further examining it (ag4; wg37; wg63; we5; eur20), illustrating a growing awareness of gender relevance to inclusive energy transformation, but lacking deeper analysis. These publications reveal a tension between a transformative change and tokenistic inclusion of gender, highlighting the need for more comprehensive, gender-responsive analyses.

Across the reviewed literature, the socio-ecological subsystem dominates (104 publications), highlighting an emerging consensus that energy transition extends beyond technological advancements, and is embedded in complex social and environmental systems (wg28 epe3; eur1; eur2; wg25; wg42). Nearly half of the publications explore

Table 2
Categories used for data extraction and organisation.

Category	Explanation
Does the publication actually assess the nexus between gender and energy transition/transformation (Yes/No)	to assess whether the publication relates to research questions and to exclude those publications that even passing the abstract screening does not match with the scope of the GLR
Does the publication's focus is on gender issues (gender as main topic, gender as subtopic, gender as a variable, without deeper consideration)	to assess whether the publication relates to research questions and to exclude publications that do not consider gender issues.
Subsystems: political economic socio-ecological technological (Bell et al., 2020)	to assess the subsystem or subsystems considered, based on the four subsystems proposed by Bell et al., 2020
Research Findings	to report the gender related results of the publication
Types of gender issue(s) addressed	to assess the type of gender issue addressed by the publication, based on the EIGE index: 1) Work; 2) Knowledge; 3) Time; 4) Health; 5) Power; 6) Safety and Trust; 7) Quality; 8) Other
Mentioned Technologies, Infrastructures and Material Systems	to assess the technology mentioned in the publication, for instance, wind, solar, hydro, coal, etc.
Which country/ies are analysed?	to assess which country or countries are assessed by the publication
Theoretical contribution	to assess how the publication contributes to the state- of - the art of gender- energy nexus.
Spatial Context	to assess the spatial context of the analysis: international, national, inner-country, peri-urban, rural
Research method	to assess the nature of the research method employed: qualitative, quantitative, mixed, theoretical
Level of analysis	to assess on what level the analysis was conducted (individual, household, institutions, sector,)
Type of analysis	to assess what kind of analysis the publication provides (casual relations, comparative, descriptive)
Sample Size (n) total	to assess the sample size of the analysis
Policy Recommendations given	to assess the presence of policy recommendations and their types.
Gender Injustices identified? (Yes/No)	to assess the presence of gender injustice within the analysis
Other Social / Vulnerability Categories interacting with Gender (Intersectional Approach)	to assess the presence of an intersectional analysis and what characteristics are included
Pathways	to assess the presence of a pathways analysis within the publication

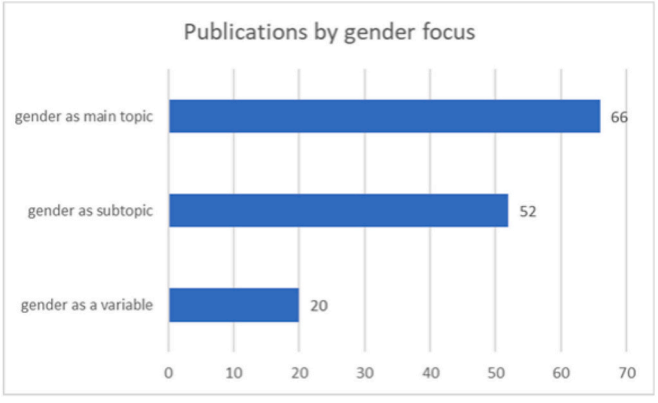


Fig. 2. Number of publications by type of gender focus.

economic (72 publications) and technological (65 publications) subsystems. Economic discussions span both individual and societal levels, often linking gender equality to economic empowerment (epg1; we4; eue6; wg43). Technology is framed not only as a driver of innovation, but also as a tool to co-produce sustained, just, and equal energy transition (wg40; wg47). These perspectives underline that just energy transitions cannot occur in a vacuum – they require attention to the social context in which technologies are applied.

In contrast, the political subsystem receives the least attention and is addressed in only 36 publications (wg45; wg42; wg49), pointing to a critical omission. Since politics shapes who defines the goals of energy transition and who has voice in decision-making, this underrepresentation risks undermining the prospects for transformative change. Applying Jasanoff's framework, stronger integration of political structures is needed to ensure that transformation co-produced by technological, economic and social advancement is supported at the decision-making level.

4.1.1. *Core gender themes*

A gender-focused analysis of the reviewed documents reveals that labour and employment constitute the most frequently discussed theme, appearing in 66 publications (Fig. 3). These discussions span both individual and structural levels, illustrating how the energy sector not only reflects but also reproduces existing gender inequalities in the labour market and broader economic structures. The reviewed literature identifies barriers to women's access to employment, challenges in career progression, limitations on entrepreneurial opportunities, gender-based occupational segregation, and pay gap (wg22; wg30; we8; we16; epg1). These issues are contextualized within broader socio-cultural and institutional frameworks that shape gendered experiences in the labour market.

Power, though less frequently discussed, emerges as a vital dimension. Attention is given to gender imbalances in formal political and economic decision-making processes, pointing to the dominance of male perspectives in energy governance. In particular, the reports point that women's contributions are often overlooked, gender-inclusive language is scarce, and female leadership remains limited (wg25; eur15). Despite limited female representation, some studies emphasize the agency of women as key actors in energy transition, positioning them as potential leaders in systemic change (we22). This shift from vulnerability to empowerment discourse may indicate a reframing of gender from a passive to an active category in energy transformation.

Other dimensions - such as knowledge (eue2), health (epg4), and allocation of time (wg51) – receive limited attention, while critical aspects like safety, trust and quality are largely absent from the reviewed publications. These gaps suggest a narrow conceptualization of the gender-energy nexus, that focuses on the most obvious and convenient dimensions. The emphasis on specific issues – such as numerical representation of women in employment and decision-making – reflects

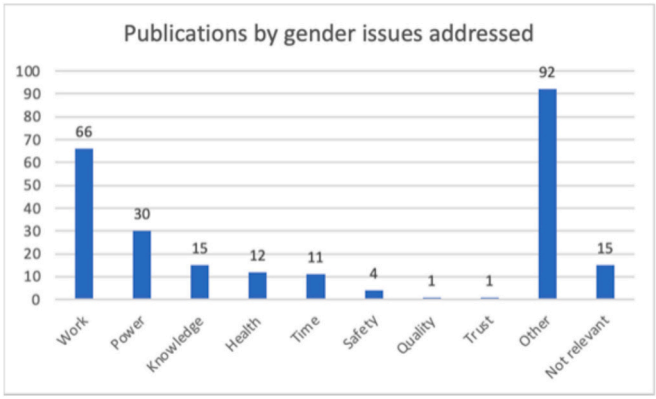


Fig. 3. Publications by gender issues addressed.

underlying normative assumption about what counts as important and potentially easy to address thanks to the existing knowledge and without substantially challenging the current economic and social systems. The marginalisation of themes like emotional labour, caregiving, or subjective well-being demonstrates how dominant epistemologies prioritise technical and economic rationalities while neglecting relational and embodied dimensions of energy use. Thus, the gender-energy nexus not only mirrors but actively shapes prevailing hierarchies of knowledge and power, pointing to a need to integrate intersectional approach in energy transformation.

The gender-energy nexus, as presented in the reviewed publications, reflects uneven patterns of engagement that align with broader epistemic and institutional biases. Although progress has been made in recognizing gender as a significant aspect of the energy transition, the integration of intersectional and systemic approaches remains limited. Only by incorporating underexplored dimensions across all subsystems, and by further addressing power asymmetries, can the co-production of knowledge and social order contribute to a just and equal energy transformation.

4.2. Technologies and the energy-gender nexus

Technology, in its materiality and symbolical meaning, plays a crucial role in shaping the current and future potential for energy production and distribution. This is influenced by past decisions and the belief in the necessity of future decisions. Existing infrastructure serves as a key determinant of the sociotechnical landscape. However, new, developing, or future technologies offer prospects for transformation within the energy system. These dynamics are intricately connected to economic viability, regulatory frameworks, public expectations, and collective imaginaries. The documents under review classify energy systems into traditional fossil fuel-based systems (coal, oil, gas) and new renewable energy-based systems (water, wind, solar, geothermal, biomass). Of the analysed documents, 51 refer to the fossil fuels system, while 144 address renewable energy sources. Emerging technologies such as nuclear and hydrogen are considered separately due to their resource limitations and the social controversies sourrounding them.

Energy technology is understood here not only in relation to energy sources, but also in connection with social needs such as heating, cooking, working, and communicating. In this context, the analysed grey literature identifies two primary challenges: pollution (primarily related to stoves) and access to energy (associated with grids). 54 documents refer to stove and 21 address grids. Both areas are gender sensitive. This is reflected in the analysis of access to various energy services, such as clean cooking technologies (we22; we18), cooling services, and transportation (epg4). Traditional gender roles have rendered women particularly vulnerable to the adverse effects of pollution from outdated cooking technologies. Furthermore, the lack of electricity significantly increases both the difficulty and the time burden of daily household tasks (wg9, ag3; wg3).

However, in documents that consider technology development or use more broadly, the gender perspective is rarely applied, or gender categories are used merely as simplistic socio-demographic descriptors without deeper analytical engagement. Documents explicitly focused on gender inequality therefore offer considerably more material to analysis than those devoted primarily to technology issues.

It is crucial to acknowledge that reports addressing the extraction of rare minerals and metals (essential to the advancement of renewable and digitalized energy systems) perpetuate long-standing challenges and social inequities in developing countries (wg42; wg66). These reports draw attention to the gender dimension of violence in this sector, and to the harmful impacts of the industry on local community, particularly on the health and well-being of women and children (wg34).

Gender is additionally proposed as a lens through which to examine the gender impact of energy solutions such as building energy efficiency. Notable differences observed between female and male workers, which are linked to diverse metabolic rates, clothing practices, and consequently distinct experiences of thermal comfort or discomfort, provide new insights into energy consumption during winter and summer periods in workplaces (wg6). Generally, gender impacts manifest in the patterns of energy consumption. Women tend to be more conscientious about environmental and climate issues and exhibit greater willingness to modify their daily practices (wg47).

Employment and the labour market constitute another critical facet of the gender-energy nexus. The grey literature review indicates that the emerging renewable resource-based energy system is more gender-sensitive than traditional fossil fuel-based systems. It is noticed that within fossil-fuel-based energy production systems, women predominantly occupy lower remunerated positions and are subjected to manifold inequalities (we8; we22).

In conclusion, gender is discussed in relation to energy technologies and resources in several ways: 1) as a factor influencing energy consumption patterns within households, workplaces, and public spaces; 2) as a dimension of injustice related to uneven access to energy services, exposure to harmful pollutants, and risks associated with outdated fuel sources; 3) as a systemic feature of injustice encompassing unequal access to STEM education, wage disparities, and the marginalized status of women in the labor market; and 4) as a context for addressing the talent gap and the untapped potential of marginalized groups within the energy sectors. However, only a limited number of reports engage in a comprehensive gender-energy analysis as a basis for conceptualising pathways toward carbon-neutral societies that also promote gender-equal social orders. One example is a policy brief examining how the transition to a green economy in sub-Saharan Africa could create new employment opportunities for women (ag1). Although the impacts of gender on energy transitions are widely acknowledged, they remain under-theorized in most reports and are infrequently integrated into technological advancement or specific policy instruments.

4.3. Spatial framings of the gender-energy nexus

Our analysis demonstrates how the gender–energy nexus is co-produced through spatial framings that structure not only how gender is understood, but also how energy systems are imagined and thus through policy documents governed. Drawing on the concept of co-production, we approach spatial scale not as a neutral backdrop but as a constitutive element in shaping both epistemic practices and institutional priorities. The reviewed literature reflects multiple levels of analysis—ranging from individual experiences to global frameworks—each producing distinct, and sometimes competing, understandings of gender in relation to energy transitions (see Fig. 4).

Importantly, understanding how spatiality is framed in grey literature allows us to trace how gender is positioned within different analytical levels and domains of action. This, in turn, shapes how specific policies, expectations, and interventions are imagined, legitimized, and enacted across energy systems globally. Through these spatial framings, grey literature does not merely reflect existing governance practices—it actively participates in defining the scales at which gender becomes visible and politically actionable.

The majority of analysed documents that specify the spatial context, focus on regions within countries (36 publications), followed by national contexts (30) and non-specified or non-relevant spatial framings (27). Less frequent are analyses at the level of regions of the world (12) or rural-urban dynamics (7). This distribution reveals a strong focus on meso-level analyses, where structural dynamics intersect with localized energy experiences. In co-productionist terms, this emphasis reflects the interplay between knowledge production (about gender roles, needs, and access) and the institutional scales at which policy and governance decisions are made. The spatial framing of gender–energy relations is not incidental—it reflects and shapes what forms of inequality are made visible, which actors are positioned as responsible, and what interventions are deemed appropriate.

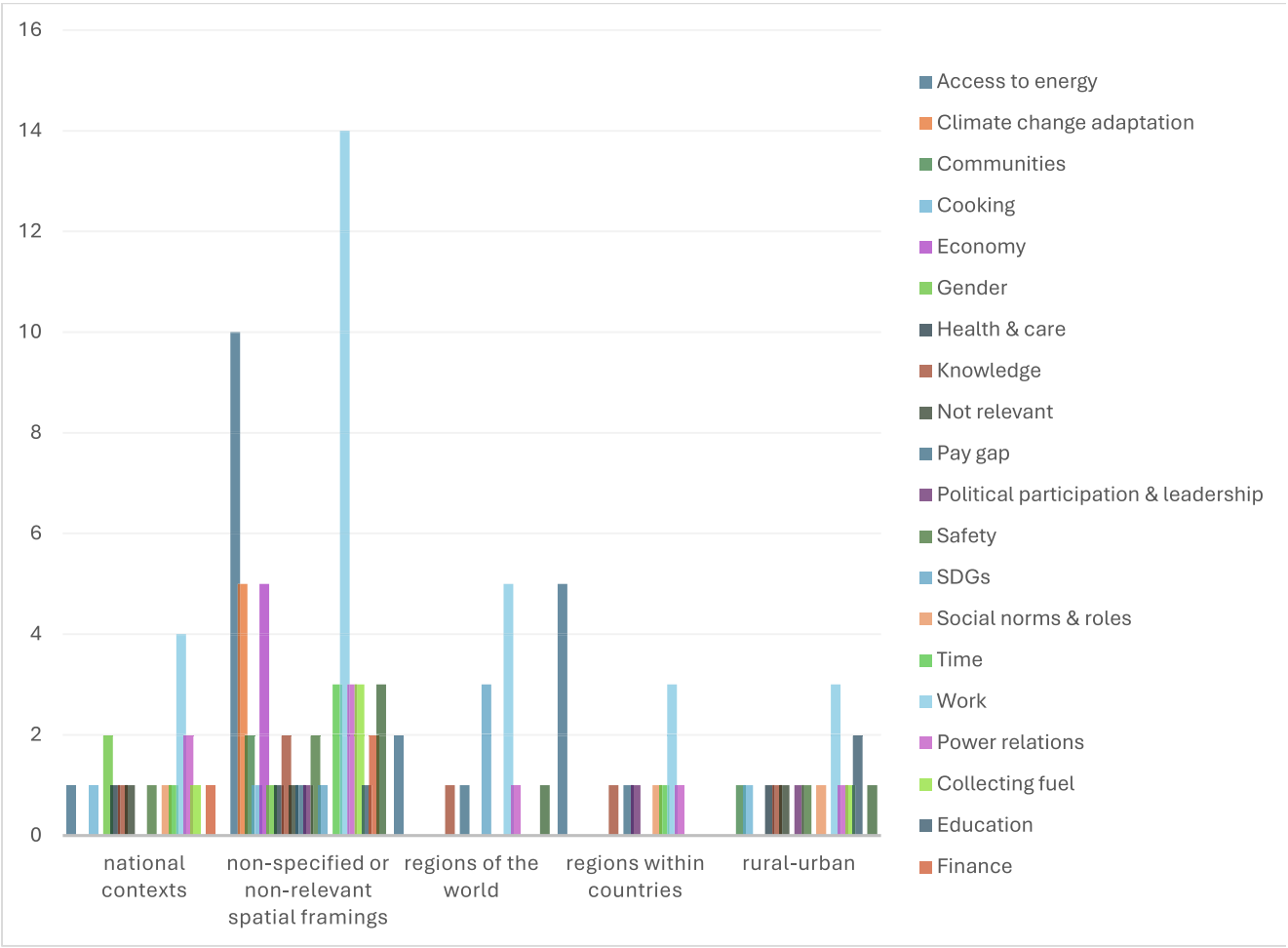


Fig. 4. Relations between types of gender issues addressed and spatial context. Categories higher than 2.

At the micro level—comprising individual and household scales—gender is often operationalized as a demographic variable rather than as a structural category. Hence, the analysis is reduced to sex-disaggregated data, typically illustrating gaps in energy access, labour, or exposure to environmental harm (wg35, wg63). This framing aligns with a more instrumental use of gender knowledge, often serving to monitor progress against global development targets like SDGs. These documents rarely engage with gender as a relational, intersectional construct, and instead reinforce a descriptive mode of knowledge that treats social differences as static variables rather than outcomes of systemic processes.

A worth noticing fact is that the household-level analyses are framed as they might offer a more situated understanding of how energy practices are embedded in traditional gender roles. These reports document how women's disproportionate responsibility for domestic energy tasks leads to cumulative disadvantages, such as time poverty, health risks, and limited educational or economic opportunities (wg63; ag1). The household, in this framing, becomes a site of co-production where material conditions (e.g. stove type, energy infrastructure) and social expectations (e.g. care responsibilities) are mutually reinforcing. Rather than isolating technical solutions from social practices, these reports begin to show how energy technologies and gender norms evolve together, shaping both lived experience and broader patterns of inequality.

Analyses at the national and regional levels represent the most theoretically expansive work in the grey literature included in this review. Reports at these scales (eur1; eur18; wg60) often examine multiple subsystems of energy transitions—economic, political, socio-ecological, and technological—and offer more intersectional framings of gender.

Studying the complex and multidimensional dynamics characterizing the processes of territory in transition, ENTRANCES embraces theoretical and methodological pluralism – a perspective in which the adoption of different scientific approaches is not considered as a problem but as an asset […]. Therefore, a multidimensional analytic framework (MAF) was adopted. The multidimensional analytic framework is articulated in five components – each relying on a set of specific concepts and methodology – and three cross-cutting elements, as shown in Fig. 2. It also shows how the components relate to the above-mentioned research foci and units of analysis (eur1).

This study is the first targeted gender assessment in Serbia's mining and energy sector, and it aims to stimulate dialogue among policy makers, companies, and other actors on how best to strengthen women's participation in mining and energy. It provides new and detailed quantitative information on women's current role in the sector and qualitative information on the key barriers to women's increased participation and success. (wg25).

Gender is treated not just as a variable, but as a lens through which to understand power relations and structural barriers. In these documents, we see a clearer engagement with how normative ideas about gender justice are entangled with governance frameworks, funding priorities, and national energy strategies. Importantly, these meso-level analyses demonstrate how institutional logics (e.g. market liberalization, just transition frameworks) shape the way gender is recognized and acted upon in policy.

At the global level, gender is most commonly framed in economic terms—access to employment, entrepreneurship, or wage equity in the energy sector (ae1; we15, we16; epg2). These reports tend to adopt a more universalizing tone, emphasizing gender equality as a "smart

economics" strategy that can enhance innovation and productivity. While this framing increases the visibility of gender in global discourse, it often narrows the scope of justice to participation and market integration, sidelining more radical or intersectional critiques. The co-production of knowledge and social order at this level reveals how global institutions privilege certain narratives of empowerment that align with existing economic models, reinforcing rather than transforming dominant paradigms.

Despite the variation across levels, a consistent theme in the literature is the identification of gender roles and norms as key barriers to equality in energy transitions (wg46; we13). Many documents highlight how entrenched cultural expectations—such as the feminization of unpaid domestic labour—continue to restrict women's access to education, mobility, and participation in the energy workforce (wg60). These norms not only shape everyday experiences but also inform the institutional assumptions embedded in energy policy and practice. In this way, the literature makes visible the recursive relationship between knowledge production and social structure: energy systems are designed and governed through assumptions about gender, which in turn reproduce the very inequalities these systems claim to address.

In conclusion, the grey literature's treatment of scale and spatial context reveals much about the co-production of gender and energy knowledge across spatial contexts. From micro-level studies that reduce gender to a monitoring variable, to meso-level analyses that embed gender in systemic critique, the spatial framing of the gender–energy nexus both reflects and reproduces institutional understandings of what matters—and for whom—in energy transitions. This layered analysis demonstrates that addressing gender inequality in energy is not only a matter of better data or inclusion, but of fundamentally rethinking how spatial, technological, and social dimensions co-evolve in the making of sustainable futures.

4.4. Areas of intervention at the gender-energy nexus

Grey literature often incorporates practical evidence and diverse perspectives that are valuable for policy-making and applied research (Yoshida et al., 2024). Accordingly, analysing recommendations related to the gendered aspects of the energy transition offers insight into the co-production of scientific knowledge and decision making within different organisations and through policy processes. Through recommendations, grey literature not only reflects current knowledge on gender-energy nexus but also participates in shaping the imaginaries of attainable futures-specifically, what gender-just energy transitions are legitimate, desired, and possible.

Within analysed publications general recommendations are interspersed with practical instructions addressed to a wide range of stakeholders, including national governments, the EU institutions and renewable energy communities, financial institutions, energy utility companies, and women's business associations. The need for full inclusion of women and men in shaping the territorial energy transition trajectories, especially in coal and carbon intensive regions has been widely recognised (we18; epg4; eur6). Those general recommendations refer to numerical inclusion of women into energy transition, both as the renewable energy workforce, energy companies' owners and managers and policy-makers (ag1; wg60, we11; a9; wg51).

More practical recommendations focus on specific areas demanding intervention, including employment, energy access, knowledge, decision-making and, to a lesser extent, culture. Many recommendations refer to the aim of equal access to managerial and engineering roles within employment structures in the energy sector as such (wg29; wg60), and specifically in renewable energy (we22; wg2; eue3; wg57). Specific measures to be implemented include quotas in construction and maintenance jobs, encouraging the establishment of women's recruitment bureaus, design, and provision of vocational training in the communities affected by the energy infrastructure projects. The perception of the energy sector as male-dominated needs to be changed through targeted communications that promote women's roles and that showcase female role models and success stories as well as through enhancing working conditions with introducing various measures for work-life balance (wg1). Similar emphasis is put on women's networking and mentoring programmes (we15; wg10; epe2).

Much attention is paid to the access to safe energy, which is seen as strongly associated with women's economic empowerment (we10, we11; wg29). In the context of African countries, Afghanistan and Uzbekistan, various financial mechanisms are proposed to support female-headed households and those with lower payment capacities. These include microfinance facilities, grants, revolving funds or credit schemes for affordable connection fee payments, reduced connection fees with longer repayment periods, subsidized connection costs, and staggered payment structures to cover the upfront costs (wg2; wg26; wg9; wg3; wg23; wg55). In the European context it is additionally advised to establish one stop-shops, providing technical assistance on renewable energy installations and (Gender) Energy Ambassadors who could operate in local communities, providing information on Do-It-Yourself measures and installing smart meters in the households (eue4; eur14, eur15).

Knowledge is seen as a prerequisite for both equitable workforce participation and access to clean renewable energy. Much attention is paid to enhancing female access to education, trainings, and targeted scholarships (wg60). Interventions should be implemented at the pre-tertiary education level, such as counselling programs targeting female students to expose them to options of STEM programs during their study at university (wg3; wg4, 2020; wg45). In the African context it is also recommended to address the differential health impacts (wg9) and carry out information campaigns on the benefits of clean and efficient cooking solutions (wg3; wg4), switching to stoves with lower emissions and changing everyday practices, such as improving ventilation to limit household members' exposure to harmful pollutants (wg31).

Recommendations on making energy decision-making gender equitable are relatively common. However, they tend to focus primarily on balancing gender representation across leadership roles, including company boardrooms (2we1; eur24, eur25), cooperatives (eue3; wg11) and political institutions, while largely overlooking the importance of integrating gender expertise and considerations into decision-making processes (epg4). When moving beyond general calls for increasing women's participation in leadership positions, reports often recommend implementing gender targets and quotas (ag3; we15). Yet these measures rarely address the underlying challenges of gender equality in the energy workplace and decision-making, including persistent institutional gender biases and masculine social and political systems. Such structural issues cannot be remedied through the "quick fix" of gender quotas (Teigen, 2021), but instead require more transformative approaches.

Interestingly, cultural norms and gender bias creating barriers for women and disadvantaged groups are addressed less frequently (epg5; we1; we8; wg60). One example is a recommendation to provide gender and inclusiveness trainings for men to demonstrate how more inclusive approaches can also benefit them and thereby reduce resistance to gender transformative interventions in renewable energy communities operating in Europe (eue4).

Irrespectively of the area of intervention, many publications conclude that the implementation of gender-responsive measures and the monitoring of their progress are hindered by a lack of data, which are described as "a driver to unlock policymaking" (we2). Thus, further research and the collection of sex-disaggregated and gender-relevant data – both quantitative and qualitative – are repeatedly recommended (wg4; eur14; epg5; wg29; wg60). At the same time, however, the need to incorporate an intersectional perspective in energy data collection, research and policy-making is acknowledged in only a few publications (eue3; eue4; epg4; eur15; wg24; wg60, wg62). Moreover, recognition of the importance of integrating gender expertise, feminist knowledge, and social movements into energy planning and policy-

making remains marginal (eur11; eur14; epg4).

5. Discussion and conclusion

Our review first reveals that gender is framed in highly uneven ways across different energy subsystems and levels of analysis. Gendered perspectives are most commonly integrated into discussions of the economic and socio-ecological dimensions of energy, while documents focused on technological systems often remain gender-blind. This suggests that gender is still largely seen as a social add-on rather than a constitutive element of the energy system. Nonetheless, where gender is discussed, the grey literature spans multiple scales—from household-level energy use to global sustainability goals—frequently invoking frameworks of gender justice to connect micro-level disparities with macro-level systemic outcomes. This multi-scalar approach illustrates co-production in action: representations of gender are shaped by, and help shape, the institutional logics that govern energy transitions at various levels, by putting emphasis and producing particular types of knowledge. This observation highlights the significant weaknesses within the energy transition framework, particularly related to the awareness and perceptions prevalent in the domain, which shape and influence other issues, such as those associated with employment.

Secondly, the intersection of gender with technology and material infrastructure remains insufficiently explored. Numerous reports implicitly regard technology as gender-neutral, thereby perpetuating a division between technical expertise and social concerns. When gender is mentioned within technical contexts—such as clean cooking stoves or off-grid energy systems—it typically involves outlining benefits for women, without conducting a deeper analysis of how these technologies are intrinsically linked to gendered norms and relations. These instances underscore the co-production of material systems and social orders: technologies are not deployed in a vacuum, but are accompanied by assumptions regarding users, roles, and needs. Nevertheless, the inconsistent integration of gender in these areas suggests that the potential for co-evolution between technological innovation and gender inclusion remains largely unrealized in most analysed material. The emphasis on health, perception, and behaviour in scientific literature only partially addresses this issue. The grey literature demonstrates the significance of technology in its material aspect in influencing individual behaviours, while also creating narratives that justify changes at regional or national levels.

Third, the analysed grey literature actively promotes particular areas of intervention, thus participating in the co-production of gendered governance. Nearly all the publications reviewed offer policy recommendations that emphasize inclusion, most commonly through efforts to increase women's participation in the energy workforce and leadership. This reflects a prevailing vision of empowerment that equates equality with access and visibility—placing more women in existing structures rather than rethinking those structures altogether. Other recommendations include improving access to clean energy technologies, expanding education and training for women in STEM, and incorporating gender perspectives into project planning. These interventions reflect institutional logics of actionability, which favour initiatives that are measurable and easily integrated into existing policy cycles, but mostly without challenging the existing power structures. The prominence of such solutions demonstrates how grey literature not only reflects global governance narratives—such as the Sustainable Development Goals and the "just transition" agenda—but also helps solidify them as dominant frameworks for energy justice.

However, our review also identifies significant limitations in how gender is framed. A major shortcoming is the tendency to treat "gender" as a synonym for "women," often without considering the power dynamics or intersecting structures of inequality that shape different women's experiences. While some reports acknowledge the need for intersectional perspectives, these remain rare. As a result, many interventions risk reproducing one-size-fits-all approaches that benefit already-advantaged women, while marginalizing others. This reductive framing is further compounded by a lack of engagement with deeper cultural and structural factors—such as the persistence of gender labour roles, educational barriers, and norms around domestic responsibility—that sustain inequality in energy systems. From a co-productionist perspective, these omissions are not accidental but reflect the institutional settings and strategic priorities of the knowledge producers. The solutions proposed are shaped by what is feasible within current governance structures, which limits their capacity to challenge the deeper roots of inequality.

Finally, our findings demonstrate how grey literature plays a pivotal role in mainstreaming gender within energy transitions. Over the past decade, these texts have helped move gender from the margins of energy policy to a more central concern. This shift is not merely a reflection of academic or activist pressure; it is itself a form of co-production, whereby institutional actors such as development banks, intergovernmental organisations, and NGOs participate in shaping what "gender equality" means in the context of energy. In doing so, grey literature transforms concepts—like gender mainstreaming or gender justice—into concrete policy instruments, such as checklists, indicators, and implementation guides. These tools then circulate across institutions, standardizing gender as a category of governance while also defining its acceptable boundaries. The co-produced nature of this process is evident: knowledge about gender and energy helps shape policy, which in turn legitimizes particular forms of knowledge.

Yet the version of gender equality that becomes institutionalized is often a narrow one, focused on inclusion within existing systems rather than structural transformation. By framing gender equity as a matter of representation and participation—rather than redistribution or deeper social change—the reviewed grey literature tends to reinforce rather than reimagine existing hierarchies. This reveals the double-edged nature of mainstreaming: it increases visibility and generates momentum but also risks entrenching limited conceptions of justice that leave underlying power structures untouched.

In summary, the grey literature on the gender–energy nexus might be seen not as a passive repository of knowledge but as an active agent in the co-production of knowledge and social order. It reflects dominant narratives about gender- sustainability nexus, inclusion, and technological progress, while also shaping how those narratives are institutionalized and acted upon. From the ways gender is framed across subsystems and levels, to the selective engagement with technologies, to the preferred forms of intervention and their conceptual limitations, each aspect of this literature reveals how energy transition is governed not only through infrastructure and policy but also through discursive and epistemic regimes.

While both academic and grey literature contribute significantly to the exploration of the gender–energy transition nexus, neither perspective offers a comprehensive account. Grey literature, although providing more context-specific insights, does not systematically capture the full spectrum of gendered experiences, particularly given the dynamic and evolving nature of the challenges involved. Concurrently, the systematic literature review by Cellini et al. (2025), while valuable in identifying major research trends, is constrained by its reliance on selected academic databases and by limited geographic diversity, focusing primarily on countries of the Global South. To better capture the complexity and local specificities of gendered aspects of energy transitions, it is therefore essential to integrate both types of sources and to include the perspectives of social actors whose experiences remain underrepresented in academic discourse.

By integrating gender within the paradigm of sustainable development, this research underscores the dual function of grey literature—both as a reflective tool and a mechanism of global sustainability governance. The uneven incorporation of gender across energy subsystems mirrors broader tensions within the sustainability agenda, particularly those between inclusivity and economic growth, and between participation and systemic transformation. Although the

discourse on gender justice and alignment with the Sustainable Development Goals signals progress toward more equitable energy transitions, gender is still too often framed as a matter of access and representation within existing social and economic systems, rather than a driver of structural change. This framing risks perpetuating existing disparities. Achieving a truly sustainable development trajectory requires recognising gender not as a supplementary aspect of energy policy but as an integral dimension of social, economic, and technological systems. In this context, the co-productive role of grey literature highlights the need to align sustainability initiatives with greater reflexivity regarding whose knowledge, needs, and values shape the transition toward a just and inclusive energy future.

Understanding grey literature as a co-productive force underscores the need for ongoing reflexivity in how gender is framed within energy systems. If the goal is to achieve just and transformative energy transitions, the knowledge practices that inform them must be critically examined and continually revised. What is at stake is not only whether gender matters, but how it matters—through which categories, narratives, and assumptions. Making these processes visible positions the co-productionist approach as a vital analytical tool for rethinking both energy governance and the knowledge systems that sustain it.

The insights derived from this review resonate strongly with the principles of transformative science (Schneidewind et al., 2016); which seeks not only to understand socio-technical systems but also to catalyse their change toward greater equity and sustainability. By revealing how gender and energy discourses are co-produced through institutional logics; governance practices; and epistemic norms; this study underscores the need for reflexivity within knowledge production itself. A transformative science perspective highlights that addressing gender in energy transitions requires moving beyond additive or representational approaches (Walk, 2024) toward reconfiguring the very frameworks through which knowledge; technology; and policy interact (Lahiri-Dutt, 2023). In this sense, examining grey literature becomes an act of transformation in its own right—making visible the assumptions, power relations, and value systems that shape what counts as legitimate knowledge and thereby opening pathways for more inclusive and justice-oriented energy futures.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Annex

Overview of reviewed publications.

Publication	Acronym	Organisation type	Organisation name	Document type
ECOWAS Centre for Renewable Energy and Energy Efficiency, *2021. Pre-feasibility Study on Business Opportunities for Women in a Changing Energy Value Chain in West Africa. Situation analysis report.* https://www.afdb.org/en/documents/pre-feasibility-study-business-opportunities-women-changing-energy-value-chain-west-africa	ae1	African: Energy Focus	ECOWAS ECREEE, AFDB, New Partnership for Africa's Development (NEPAD) Infrastructure Project, Preparatory Facility (IPPF) Special Fund	REPORT
African Development Bank & UN Women, 2021. *Green Jobs for Women in Africa: Opportunities and Policy Interventions*. AfDB. https://www.afdb.org/sites/default/files/documents/publications/un_woman_green_jobs_policy_brief_eng_0.pdf	ag1	African: General Focus	AFDB and UN WOMEN	Policy brief
African Development Bank and United Nations Economic Commission for Africa, 2020a. *Africa Gender Index Report 2019. What the 2019 Africa Gender Index tells us about gender equality, and how can it be achieved.* African Development Bank Group. https://www.afdb.org/sites/default/files/documents/publications/africa_gender_index_report_2019_-_analytical_report.pdf	ag2	African: General Focus	AFDB, Gender, Poverty and Social Development Policy Division at the UN Economic Commission for Africa	Report
African Development Bank and United Nations Economic Commission for Africa, 2020b. *Methodological & Statistical Report. Bridging the Gender Gaps: The Africa Gender Index Report 2019*. African Development Bank Group. https://www.afdb.org/en/documents/africa-gender-index-2019-methodological-and-statistical-report	ag3	African: General Focus	AFDB, Gender, Poverty and Social Development Policy Division at the UN Economic Commission for Africa	Report
African Development Bank Group, 2019. *Clean energy to power Africa's future.* https://www.afdb.org/en/documents/clean-energy-power-africas-future	ag4	African: General Focus	AFDB	Brief
African Development Bank Group, 2022. *Country Focus Report 2022. Supporting climate resilience and a just energy transition in Ghana.* https://www.afdb.org/en/documents/country-focus-report-2022-supporting-climate-resilience-and-just-energy-transition-ghana	ag5	African: General Focus	AFDB	Report

(*continued on next page*)

(*continued*)

Publication	Acro-nym	Organisation type	Organisation name	Document type
African Development Bank Group, 2022. *Just Transition in a Renewable Energy Rich Environment - Potential Role of Green Hydrogen*. https://www.afdb.org/en/documents/just-transition-renewable-energy-riche-environment-potential-role-green-hydrogen	ag6	African: General Focus	AFDB	Report
African Development Bank, 2020. *The African Development Bank – Gender Marker System (GMS)*. AfDB. https://www.afdb.org/en/documents/african-development-bank-gender-marker-system-gms	ag7	African: General Focus	AFDB	Report
Moungar, V., 2019. *Mainstreaming Gender in Our Climate Action for Sustainable Impact*. African Development Bank.https://www.afdb.org/en/documents/mainstreaming-gender-our-climate-action-sustainable-impact	ag8	African: General Focus	AFDB	Brief
INTRASOFT International S.A., PLANET S.A., White Research, Valeu Consulting, Navigant, ENTSO-G, 2021. Innovative Actions and Strategies to Boost Public Awareness, Trust and Acceptance of Trans-European Energy Infrastructure Projects: Final Report. European Commission.	eue1	EU: Energy Initatives	EU Commision DG Energy	Report
Korteweg, L., Beznea, A., Yearwood, J., Guevara Opinska, L., Eklund, L., 2022. Case Study Analysis of Regions in Transition. Final Report. Publications Office of the European Union, Luxembourg	eue2	EU: Energy Initatives	EU Commision DG Energy	Report
Clancy, J., & Feenstra, M., 2019. *Women, gender equality and the energy transition in the EU*. Publications Office of the European Union. https://www.europarl.europa.eu/RegData/etudes/STUD/2019/608867/IPOL_STU(2019)608867_EN.pdf	eue3	EU: Energy Initiatives	European Parliament's Committee on Women's Rights and Gender Equality	Report
Clancy, J., Kustova, I., Elkerbout, M., & Michael, K., 2022. *The Gender Dimension and Impact of the FITfor55 Package*. https://www.europarl.europa.eu/RegData/etudes/STUD/2022/736899/IPOL_STU(2022)736899_EN.pdf	eue4	EU: Energy Initiatives	European Parliament's Committee on Women's Rights and Gender Equality (FEMM).	Report
European Commission, 2020. *Striving for a Union of equality: The gender equality strategy 2020–2025*. Directorate-General for Communication. https://op.europa.eu/en/publication-detail/-/publication/84bd5dd5-7879-11ea-a07e-01aa75ed71a1/language-en?utm_source=chatgpt.com	eue5	EU: Energy Initiatives	EU Commission	Strategy
European Commission: Directorate-General for Energy, E3Modelling, Tractebel Impact, Chaumont, S., Charalampidis, I., Demkova, D., 2021. *ASSET study on collection of gender-disaggregated data on the employment and participation of women and men in the energy sector*, Publications Office. https://data.europa.eu/doi/10.2833/888421	eue6	EU: Energy Initiatives	EU Commission Directorate-General for Energy, Directorate for Energy Policy: Strategy and Coordination Equality Network	Report
Barrett, T., 2022. *Lusatia Case Study Report (Deliverable D3.2)*. ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels.	eur1	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Barrett, T., Bréau, M., Caiati, G., Healy, A., Heinisch, K., Holtemöller, O., Marshall, A., Norris, L., Škobla, D., Singh, N., Schult, C., 2022. South Wales Region Case Study Report (Deliverable D4.6). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D4.6-South-Wales-Case-Study-Report.pdf	eur2	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Bartl, W., Heinisch, K., Holtemöller, O., Schult, C., 2022. *Rhineland Case Study Report (Deliverable D3.3)*. ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels.	eur3	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Bartl, W., Schult, C., Heinisch, K., Holtemöller, O., 2022. Central Germany Case Study Report (Deliverable D3.4). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D3.4-CentralGermany-Case-Study-Report.pdf	eur4	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Birgi, O.G., Fuhrmann, A., Habersbrunner, K., Stock, A., 2021. Gender and energy poverty: Facts and arguments. EmpowerMed Project. https://www.empowermed.eu/wp-content/uploads/2021/05/2104.Empowermed-Energy_Poverty_and_gender.pdf	eur5	EU: research	EmpowerMed (Empowering Women to Take Action Against Energy Poverty)	report
Bréau, M., Noreña Ospina, M., Strebel, A., Baumann, J., 2023. Gender-based Report (Deliverable 5.2). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2023/04/Deliverable-5.2-Gender-based-report_Final.pdf	eur6	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Caiati, G., Quinti, G. Bassano, C., Deiana, P., 2022. Sulcis Case Study Report (Deliverable D3.6). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2023/06/D3.6-Sulcis-Case-Study-Report.pdf	eur7	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society):	Project Deliverable
Caiati, G., Quinti, G., Cacace, M., Feudo, F., Garcia Mira, R., Singh Garha, N., González Laxe, F., Rey Vizoso, F.J., Filčák, R., Škobla, D., Holtemöller, O., Schult, C., Heinisch, K., Wolfram, M., Barrett, T., Knippschild, R., Rühlemann, A., Kuschan, M., Noreña, M., De Luca, E., Spiesberger, M., Klöckner, C., Haley, A., Boncu, S., Holman, A., 2021. Report on Multi-dimensional Key Factors, Dynamics and Patterns (Deliverable	eur8	EU: Research	ENTRANCES (Energy TRANsitions from	Project Deliverable

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D1.2). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2021/06/D1.2.-Report-on-Multidimensional-Key-Factors.pdf			Coal and carbon: Effects on Society)	
Caiati, G.,M., Feudo, F. (K&I); Garcia Mira, R., Singh Gartha, N., González Laxe, F., Rey Vizoso, F.J. (UDC); Filčák, R., Škobla, D. (CSPS); Holtemöller, O., Schult, C., Heinisch, K. (IWH); Wolfram, M., Barrett, T., Knippschild, R. (IOER); Rühlemann, A., Kuschan, M., Norena, M. (WECF); De Luca, E. (ENEA); Spiesberger, M. (ZSI); Klöckner, C. (NTNU); Haley, A. (UC); Boncu, S., Holman, A. (UAIC), 2022. *Jiu Valley Region Case Study Report (Deliverable D3.5)*. ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/project-deliverables/	eur9	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society):	Project Deliverable
Cerone, N., Cotroneo, R., 2022. Brindisi Region Case Study Report (Deliverable D4.1). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D4.1-Brindisi-Case-Study-Report.pdf	eur10	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
EGC. (2023). Replication Guide and Toolkit – First Version. W4RES Project. https://cordis.europa.eu/project/id/952874/results	eur11	EU: Research	W4RES (Women for market uptake of renewable heating and cooling)	Report Deliverable
Filčák, R., Škobla, D., Malová, D., Pollák, M., 2022. Horná Nitra Case Study Report (Deliverable D3.7). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D3.7-HORNA-NITRA-Case-Study-Report.pdf	eur12	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Frantál, B., 2020. Demographic trends and challenges in gender, migration and youth – A literature review. TIPPING.plus H2020 Project. https://cordis.europa.eu/project/id/884565/results	eur13	EU: Research	TIPPING.plus (Enabling Positive Tipping Points towards clean-energy transitions in Coal and Carbon Intensive Regions)	Report Deliverable
Groneweg, K., Habersbrunner, K., Stock, A., 2023. Policy recommendations for gender-just policies to reduce energy poverty. EmpowerMed Project (Horizon 2020, Grant Agreement No. 847052). https://www.wecf.org/de/wp-content/uploads/2023/04/Gender-just-policy-recommendations-EmpowerMed-2023.pdf	eur14	EU: Research	EmpowerMed (Empowering Women to Take Action Against Energy Poverty)	Policy recommendation
Habersbrunner, K., Martschew, E.-C., 2020. Report on gender aspects of existing financial schemes for energy poverty measures. WECF. https://www.wecf.org/report-on-gender-aspects-of-existing-financial-schemes-for-energy-poverty-measures/	eur15	EU: Research	EmpowerMed (Empowering Women to Take Action Against Energy Poverty)	Report
Komorowska, A., Hubert, W., Kowalik, W., Kryzia, D., Gawlik, L., Pepłowska, M., Mokrzycki, E., Uberman, R., Mirowski, T., 2022. Krakow Metropolitan Area Case Study Report (Deliverable D4.2). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D4.2-KMA-Case-Study-Report.pdf	eur16	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Komorowska, A., Hubert, W., Kowalik, W., Kryzia, D., Peplowska, M., Gawlik, L., Mokrzycki, E., Mirowski, T., Uberman, R., 2022. *Silesia Region Case Study Report (Deliverable D3.1)*. ENTRANCES H2020 Project, Katowice: IGSMiE PAN / ENTRANCES Consortium.	eur17	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
Singh Garha, N., Garcia Mira, R.A., Fernandez Preto, M., Espiñeira, E.M., 2022. A Coruña Region Case Study Report (Deliverable D4.3). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D4.3-ACoruna-Case-Study-Report.pdf	eur18	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
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Standal, K., Aakre, S., Alonso, I., Azevedo, I., Wnuk, R., Di Nucci, M. R., Krug, M., Kudrenickis, I., & Maleki-Dizaji, P., 2022. *COME RES Deliverable 2.3: Synthesis Report of Case-Studies on Drivers and Barriers in 5 Selected Target Regions*. Zenodo. https://come-res.eu/fileadmin/user_upload/Resources/Deliverables/COME_RES_D2_3_synthesis_case_studies_of_drivers_and_barriers.pdf	eur20	EU: Research	COME RES (Community Energy for the Uptake of Renewables in the Electricity Sector)	Working Paper
Vilhelmsen, K., Fjællingsdal, K.S., Nayum, A., 2022. Stavanger Region Case Study Report (Deliverable D4.5). ENTRANCES H2020 Project. ENTRANCES Consortium, Brussels. https://entrancesproject.eu/wp-content/uploads/2022/12/D4.5-Stavanger-Case-Study-Report.pdf	eur21	EU: Research	ENTRANCES (Energy TRANsitions from Coal and carbon: Effects on Society)	Project Deliverable
W4RES, 2021. Market Uptake Support Measures – First Version. W4RES Project. Retrieved from https://cordis.europa.eu/project/id/952874/results	eur22	EU: Research	W4RES (Women for market uptake of renewable	Report Deliverable

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Publication	Acro-nym	Organisation type	Organisation name	Document type
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WECF, 2021. Deliverable 1.3: Case Studies of Women Leading RHC Market Uptake. W4RES Deliverable 1.3. Women Engage for a Common Future (WECF e.V.), Munich. https://cordis.europa.eu/project/id/952874/results	eur24	EU: Research	W4RES (Women for market uptake of renewable heating and cooling)	Report Deliverable
WECF, 2022. Capacity Building Program for Women Empowerment in RHC Sector. W4RES Deliverable 4.2. Women Engage for a Common Future (WECF e.V.), Munich. https://cordis.europa.eu/project/id/952874/results	eur25	EU: Research	W4RES (Women for market uptake of renewable heating and cooling)	Report Deliverable
White Research, 2022. Report on regional hackathons. W4RES H2020 Project. https://cordis.europa.eu/project/id/952874/results	eur26	EU: Research	W4RES (Women for market uptake of renewable heating and cooling)	Report Deliverable
White Research, 2021. Needs, Perceptions and Challenges in the Renewable Heating and Cooling (RHC) Landscape: Evidence from 8 Regions. W4RES Deliverable 1.2. White Research, Brussels. https://cordis.europa.eu/project/id/952874/results	eur27	EU: Research	W4RES (Women for market uptake of renewable heating and cooling)	Report Deliverable
Global Women's Network for the Energy Transition (GWNET), 2019. *Women for Sustainable Energy: Strategies to Foster Women's Talent for Transformational Change*. GWNET. https://www.globalwomennet.org/women-for-sustainable-energy/	epg1	Policy making in Europe: Gender focus	Global Women's Network for the Energy Transition	Report
Ban, M., Brajković, J., Buzarovski, S., Lončarević, Š., Maras, J., Robić, S., Stupin, K., Šantalab, D., Toš, A., Zidar, M., 2021. Study on Addressing Energy Poverty in the Energy Community Contracting Parties. DOOR & EIHP. https://www.eihp.hr/wp-content/uploads/2021/12/DOOREIHP_poverty_122021-1.pdf	epe1	Policy-making in Europe: Energy Focus	Other: Energy Community	Report
Giner-Reichl, I., & van Veldhuizen, M., 2023. *Europe's Energy Transition: Women's Power in Solving the Labour Bottleneck – Employment Opportunities and Requirements for Low-carbon Job Markets*. Friedrich-Ebert-Stiftung. https://library.fes.de/pdf-files/bueros/bruessel/20418.pdf	epe2	Policy-making in Europe: Energy Focus	Friedrich-Ebert-STiftung (FES Just Climate), Global Women's Network for the Energy Transition	report
Grabbe, H., Potočnik, J., & Dixson-Declève, S., 2022. *International System Change Compass: The global implications of achieving the European Green Deal*. SYSTEMIQ, The Club of Rome, Open Society European Policy Institute. https://www.opensocietyfoundations.org/uploads/00519f1b-7877-4ed0-ba64-dd34cd81e410/international-system-change-compass-report-20220526.pdf	epe3	Policy-making in Europe: Energy Focus	Other: SYSTEMIQ, The Club of Rome, and the Open Society European Policy Institute	Report
Beck, Z., & Pánczél, A., 2018. Women in Energy: Gender Diversity in the CEE-SEE Energy Sector. Women in Energy. https://www.womeninenergy.eu/wp-content/uploads/2018/12/Women_in_Energy_in_the_CEE-SEE_Region_Dec2018_final.pdf	epg2	Policy-making in Europe: Gender Focus	Women in Energy	Report
Beck, Z., & Pánczél, A., 2023. *Women in Energy 2.0: Gender Diversity in the CEE-SEE Energy Sector*. Boston Consulting Group. https://www.womeninenergy.eu/wp-content/uploads/2023/06/BCG_WONY_Women_in_Energy_Study_2023_Final.pdf	epg3	Policy-making in Europe: Gender Focus	Women in Energy	Report
European Environmental Bureau (EEB) & Women Engage for a Common Future (WECF), 2021. *Why the European Green Deal needs ecofeminism: Moving from gender-blind to gender-transformative environmental policies*. https://www.wecf.org/wp-content/uploads/2021/07/Report_Green-Deal-Gender.pdf	epg4	Policy-making in Europe: Gender Focus	European Environmental Bureau, Women Engage for a Common Future (WECF)	Report
Ngum, S., & Kim, L., 2023. *Powering a Gender-Just Energy Transition*. Green Growth Knowledge Partnership. https://gggi.org/report/powering-a-gender-just-energy-transition/	epg5	Policy-making in Europe: Gender Focus	Other: The Green Growth Knowledge Partnership (GGKP) Expert Group on Gender	Working Paper
BMWK (Federal Ministry for Economic Affairs and Energy, Germany), 2022. *G7 Report on Gender Equality and Diversity in the Energy Sector*. https://www.bmwk.de/Redaktion/EN/Artikel/Energy/g7-report-on-gender-equality-and-diversity-in-the-energy-sector.html	we1	Worldwide: Energy Focus	Other: Federal Ministry fo Economic affairs and Climate Action (G7)	Report
Global Wind Energy Council (GWEC), 2021. *Submission of Views on: Mainstreaming Gender Considerations into a National Climate Change Adaptation Framework*. https://www4.unfccc.int/sites/SubmissionsStaging/Documents/201907151427—GWEC-GWNET%20-%20UNFCCC%20Submission%20on%20Gender%20Mainstreaming%20-%20Final.pdf	we2	Worldwide: Energy Focus	Global Wind Energy Council (GWEC), Global Women's Network for the Energy Transition (GWNET)	Submission of views

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Publication	Acro-nym	Organisation type	Organisation name	Document type
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International Energy Agency (IEA), 2019. *Women working in the rooftop solar sector: A look at India's transition to clean energy*. IEA, Paris. https://www.iea.org/reports/women-working-in-the-rooftop-solar-sector	we4	Worldwide: Energy Focus	IEA	Report
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International Energy Agency (IEA), 2022. *How Governments Support Clean Energy Start-ups: Insights from selected approaches around the world*. IEA, Paris. https://www.iea.org/articles/women-in-cleantech-challenge	we6	Worldwide: Energy Focus	IEA	Report
International Energy Agency (IEA), 2022. *Skills Development and Inclusivity for Clean Energy Transitions*. IEA, Paris. https://iea.blob.core.windows.net/assets/953c5393-2c5b-4746-bf8e-016332380221/Skillsdevelopmentandinclusivityforcleanenergytransitions.pdf	we7	Worldwide: Energy Focus	IEA	Report
International Energy Agency (IEA), 2022. *Understanding gender gaps in wages, employment and career trajectories in the energy sector*. International Energy Agency. https://www.iea.org/articles/understanding-gender-gaps-in-wages-employment-and-career-trajectories-in-the-energy-sector	we8	Worldwide: Energy Focus	IEA	Report
International Energy Agency (IEA), International Renewable Energy Agency (IRENA), United Nations Statistics Division (UNSD), World Bank, & World Health Organisation (WHO), 2021. *Tracking SDG 7: The Energy Progress Report 2021*. World Bank. https://www.irena.org/Publications/2021/Jun/Tracking-SDG-7-2021	we9	Worldwide: Energy Focus	IRENA, IEA, UNSD, WB, WHO	Report
International Energy Agency (IEA), International Renewable Energy Agency (IRENA), United Nations Statistics Division (UNSD), World Bank, & World Health Organisation (WHO), 2022. *Tracking SDG 7: The Energy Progress Report 2022*. World Bank. https://www.irena.org/Publications/2022/Jun/Tracking-SDG-7-2022	we10	Worldwide: Energy Focus	IRENA, IEA, UNSD, WB, WHO	Report
International Energy Agency (IEA), IRENA, UNSD, World Bank, WHO, 2023. *Tracking SDG 7: The Energy Progress Report*. World Bank, Washington DC. https://digitallibrary.un.org/record/4012885?ln=en&v=pdf	we11	Worldwide: Energy focus	IEA/ IRENA	Report
International Renewable Energy Agency (IRENA) & SELCO Foundation, 2022. *Fostering livelihoods with decentralised renewable energy: An ecosystems approach*. International Renewable Energy Agency, Abu Dhabi. https://www.irena.org/Publications/2022/Jan/Fostering-Livelihoods-with-Decentralised-Renewable-Energy	we12	Worldwide: Energy Focus	IRENA	Report
International Renewable Energy Agency (IRENA) Coalition for Action, 2021. *Community Energy Toolkit: Best practices for broadening the ownership of renewables*. International Renewable Energy Agency, Abu Dhabi. https://www.irena.org/-/media/Files/IRENA/Coalition-for-Action/IRENA_Coalition_Energy_Toolkit_2021.pdf?rev=7f4fbb8e2ffe4d208e7b70e4a768435d	we13	Worldwide: Energy Focus	IRENA	White paper
International Renewable Energy Agency (IRENA) Sustainable Energy Jobs Working Group & COBENEFITS, 2021. *Renewable energy, employment opportunities and skill requirements: Socio-economic assessment tools, key findings and expert contacts*. https://justtransitionforall.com/reports-new/renewable-energy-employment-opportunities-and-skill-requirements-socio-economic-assessment-tools-key-findings-and-expert-contacts-draft/	we14	Worldwide: Energy Focus	IRENA Sustainable Energy Jobs Working Group and COBENEFITS	Factsheet
International Renewable Energy Agency (IRENA), 2019. *Renewable energy: A gender perspective*. International Renewable Energy Agency, Abu Dhabi. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Jan/IRENA_Gender_perspective_2019.pdf?rev=bed1c40882e54e4da21002e3e1939e3d	we15	Worldwide: Energy Focus	IRENA	Report
International Renewable Energy Agency (IRENA), 2020. *Wind energy: A gender perspective*. International Renewable Energy Agency, Abu Dhabi. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Jan/IRENA_Wind_gender_2020.pdf?rev=270b62baad3c40a5b289a4f47eb8c5a9	we16	Worldwide: Energy Focus	IRENA	Report
International Renewable Energy Agency (IRENA), 2022. *Solar PV: A gender perspective*. International Renewable Energy Agency. https://www.irena.org/Publications/2022/Sep/Solar-PV-A-gender-perspective	we17	Worldwide: Energy Focus	IRENA	Report
Sustainable Energy for All, 2018b. *Levers of Change: How Global Trends Impact Gender Equality and Social Inclusion in Access to Sustainable Energy*. https://www.seforall.org/publications/levers-of-change-how-global-trends-impact-gender-equality-and-social-inclusion-in	we18	Worldwide: Energy Focus	SE4ALL	Report
Sustainable Energy for All, 2020. *Energizing Finance: Understanding the Landscape*. https://www.seforall.org/publications/energizing-finance-understanding-the-landscape-2020/energizing-finance-financing-for-gender-focused-energy-access https://www.seforall.org/system/files/2020-12/EF-2020-UL-Gender-SEforALL.pdf	we19	Worldwide: Energy Focus	SE4ALL	Brief
Sustainable Energy for All, 2021. *Cooling for All and Gender: Towards Inclusive, Sustainable Cooling Solutions*. https://www.seforall.org/publications/cooling-for-all-and-gender	we20	Worldwide: Energy Focus	SE4ALL	Brief
World Health Organisation (WHO), 2023. *Energizing health: Accelerating electricity access in health-care facilities*. World Health Organisation. https://www.who.int/publications-detail-redirect/9789240066960	we21	Worldwide: Energy Focus	IRENA, WHO	Report
Sustainable Energy for All, 2018a. *Evaluating Government and Business Landscapes on Women's Empowerment in Sustainable Energy*. https://www.seforall.org/data-and-evidence/evaluating-government-and-business-landscapes-on-womens-empowerment-in-sustainable	we22	Worldwide: Energy Focus	SE4ALL	Report
Ahmad, A., Kantarjian, L., El Ghali, H., Maier, E., Constant, S., 2019. Shedding Light on Female Talent in Lebanon's Energy Sector. Energy Sector Management Assistance Program, World Bank, Washington, D·C.	wg1	Worldwide: General Focus	WB	Policy brief
Angelou, N., Roy, S., 2019. Integrating Gender and Social Dimensions into Energy Interventions in Afghanistan. World Bank, Washington, D·C.	wg2	Worldwide: General Focus	WB	Policy brief
Brutinel, M. et al., 2020. Niger – Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework. World Bank, Washington, D.C.	wg3	Worldwide: General Focus	WB	Report
Brutinel, M., Wang, Y., Koo, B.B., Portale, E., Rysankova, D., 2019. São Tomé and Príncipe – Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework. World Bank, Washington, D.C.	wg4	Worldwide: General Focus	WB	Report
Cabraal, A., Ward, W.A., Bogach, V.S., Jain, A., 2021. Living in the Light: The Bangladesh Solar Home Systems Story. World Bank, Washington, D.C.	wg5	Worldwide: General Focus	WB	Report

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Publication	Acro-nym	Organisation type	Organisation name	Document type
Canpolat, E., Casabonne, U., 2021. Gender Differences in Behavior and Perceptions of Energy Efficiency in Public Buildings in Turkey. World Bank, Washington, D.C.	wg6	Worldwide: General Focus	WB	Report
Dazé, A., & Hunter, C., 2022. Gender-responsive National Adaptation Plan (NAP) processes: Progress and promising examples (NAP Global Network synthesis report 2021–2022). International Institute for Sustainable Development. https://napglobalnetwork.org/wp-content/uploads/2022/08/napgn-en-2022-gender-nap-synthesis-report.pdf	wg7	Worldwide: General focus	Other: NAP Global Network Secretariat c/o International Institute for Sustainable Development (IISD)	Annual report
Demmelbauer, V., Noreña Ospina, M., Baumann, J., Habersbrunner, K., 2023. Foundation Plan for Establishing a Gender-Just Energy Cooperative in Ethiopia and Uganda. Women Engage for a Common Future (WECF e.V.). https://www.wecf.org/foundation-plan-for-establishing-a-gender-just-energy-cooperative-in-ethiopia-and-uganda/	wg8	Worldwide: General Focus	WECF - Women Engage for a Common Future	Report
Dubey, S., Adovor, E., Rysankova, D., Koo, B.B., 2020. Kenya – Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework. World Bank, Washington, D.C.	wg9	Worldwide: General Focus	WB	Report
Duval, A. 2020. *Sectoral Assessment of Women's Entrepreneurship Development in the Agriculture and Renewable Energy Sectors in Somalia.* ILO. https://www.ilo.org/publications/sectoral-assessment-womens-entrepreneurship-development-agriculture-and	wg10	Worldwide: General Focus	ILO	Report
Eichermüller, J., Furlan, M., Habersbrunner, K., Kordić, Z., 2018. Energy cooperatives: Comparative analysis in Eastern Partnership countries and Western Balkans. WECF & ZEZ. https://www.wecf.org/comparative-study-on-energy-cooperatives-in-eastern-europe-and-the-western-balkans/	wg11	Worldwide: General Focus	WECF Women Engage for a Common Future (WECF), Zelena energetska zadruga (ZEZ)	Report
Energy Sector Management Assistance Program, 2019. Gender Equality in the Geothermal Energy Sector: Road to Sustainability. ESMAP Knowledge Series No. 028/19, World Bank, Washington, D.C.	wg12	Worldwide: General Focus	WB	Report
Energy Sector Management Assistance Program, 2019. Uganda Clean Cooking Behavioral Diagnostic. ESMAP Paper, World Bank, Washington, D.C.	wg13	Worldwide: General Focus	WB	Report
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Energy Sector Management Assistance Program, 2021. What Drives the Transition to Modern Energy Cooking Services?: A Systematic Review of the Evidence. World Bank, Washington, D.C.	wg15	Worldwide: General Focus	WB	Report
ESMAP, 2020. The State of Access to Modern Energy Cooking Services. World Bank Group, Washington, D.C.	wg16	Worldwide: General Focus	WB	Report
International Finance Corporation, 2022. The Business Case for Gender-Responsive Climate-Smart Mining. Washington, D.C.	wg17	Worldwide: General Focus	WB	Report
International Labour Organisation, 2018. *Skills for Green Jobs in Bangladesh.* https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_emp/%40ifp_skills/documents/publication/wcms_706892.pdf	wg18	Worldwide: General Focus	ILO	Report
International Labour Organisation, 2020. *Market Systems Analysis for Skills in the Renewable Energy and Energy Efficiency Sub-sectors in Zambia.* ILO, Zambia. https://www.ilo.org/publications/market-systems-analysis-skills-renewable-energy-and-energy-efficiency-sub	wg19	Worldwide: General Focus	ILO	Report
International Labour Organisation, 2022. *A just energy transition in Southeast Asia: The impact of coal phase-out on jobs.* ILO, Thailand. https://www.ilo.org/publications/just-energy-transition-southeast-asia-impact-coal-phase-out-jobs	wg20	Worldwide: General Focus	ILO	Report
International Labour Organisation, United Nations Entity for Gender Equality and the Empowerment of Women, 2021. *Empowering Women at Work. Government Laws and Policies for Gender Equality.*ILO, Geneva. https://www.ilo.org/wcmsp5/groups/public/—ed_emp/—emp_ent/—multi/documents/publication/wcms_773233.pdf	wg21	Worldwide: General Focus	ILO	Report
International Renewable Energy Agency, International Labour Organisation, 2021. *Renewable Energy and Jobs. Annual Review 2021. International Renewable Energy Agency, International Labour Organisation, Abu Dhabi, Geneva. https://www.ilo.org/publications/renewable-energy-and-jobs-annual-review-2021*	wg22	Worldwide: General Focus	ILO	Report
Koo, B.B., Rysankova, D., Portale, E., Angelou, N., Keller, S., Padam, G., 2018. Rwanda – Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework. World Bank, Washington, D.C.	wg23	Worldwide: General Focus	WB	Report
Kuschan, M., Burghard, U., Groneweg, K., Strebel, A., 2022. *Is the German energy transition perceived as gender- and socially-just?* Fraunhofer Institute for Systems and Innovation Research ISI (Ed.), Karlsruhe.	wg24	Worldwide: General Focus	WECF e.V. Deutschland; Fraunhofer Institute for Systems and Innovation Research ISI	Report
Lukić, J., Faria, M.M., Perks, R.B., 2022. Exploring Opportunities for Gender Diversity in the Mining and Energy Sector in Serbia. World Bank Group, Washington, D.C.	wg25	Worldwide: General Focus	WB	Report
Luzi, L., Lin, Y., Koo, B.B., Rysankova, D., Portale, E., 2019. Zambia – Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework. Energy Sector Management Assistance Program, World Bank, Washington, D.C.	wg26	Worldwide: General Focus	WB	Report
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Manandhar, M., Hawkes, S., Buse, K., Nosrati, E., & Magar, V., 2018. *Gender, health and the 2030 agenda for sustainable development.* Bulletin of the World Health Organisation, 96(9), 644–653.	wg28	Worldwide: General Focus	WHO	Bulletin
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Publication	Acro-nym	Organisation type	Organisation name	Document type
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United Nations Economic and Social Commission for Western Asia, 2020. *The Regional Initiative for Promoting Small-Scale Renewable Energy Applications in Rural Areas of the Arab Region (Regend): Study on Gender Mainstreaming, Social Inclusion, Human Rights Process and Outcomes of Access to Energy in Targeted Local Community in Tunisia*. https://digitallibrary.un.org/record/3952415?ln=en&v=pdf	wg39	Worldwide: General Focus	UN/ESCWA	Report
United Nations Economic and Social Council, 2018. *Report on Energy Transition in Asia and the Pacific: Pathways for Ensuring Access To Affordable, Reliable, Sustainable and Modern Energy for All: Note by the Secretariat*. https://digitallibrary.un.org/record/3882236?ln=en&v=pdf	wg40	Worldwide: General Focus	UN	Report
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United Nations, 2022. *Management response to the evaluation of UNDP support to energy access and transition*. https://digitallibrary.un.org/record/3952460?ln=en&v=pdf	wg52	Worldwide: General Focus	UN	Management response/ Policy document

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Publication	Acro-nym	Organisation type	Organisation name	Document type
World Bank Group, 2018. Incentivizing a Sustainable Clean Cooking Market: Lessons from a Results-Based Financing Pilot in Indonesia. World Bank, Washington, D.C.	wg53	Worldwide: General Focus	WB	Report
World Bank Group, 2019. Key Entry Points for Gender Equity in Energy Access, Energy Policy, Renewable Energy, and Transmission and Distribution. World Bank, Washington, D.C.	wg54	Worldwide: General Focus	WB	Practical guidance
World Bank, 2019. Energy Vulnerability in Female-headed Households: Findings from the Listening to Citizens of Uzbekistan Survey. World Bank, Washington, D.C	wg55	Worldwide: General Focus	WB	Report
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World Bank, 2020. Design of Financial Support and Capacity-Building Program for Rooftop Solar Photovoltaic in Turkey. World Bank, Washington, D.C.	wg57	Worldwide: General Focus	WB	Summary note
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World Bank, 2022. A Sure Path to Renewable Energy: Maximizing Socioeconomic Benefits Triggered by Renewables. World Bank, Washington, D.C.	wg60	Worldwide: General Focus	WB	Technical guide
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Zhang, Y., Adams, N., Pemberton-Pigott, C.R., Pinto, A.N., Wu, J., Barnes, D.F., Tuntivate, V.T., Durix, L., 2020. Unlocking Clean Cooking Pathways: A Practitioner's Keys to Progress. World Bank Group, Washington, D.C	wg65	Worldwide: General Focus	WB	Report
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Data availability

A list of all reviewed publications has been shared in the Annex

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